KCC CODE CLUB · GAME MAKERS

Week 6

Broadcasts and Game Over

Week 6 of 8

Name _____

Today's big idea

Sprites can't read each other's code. But one sprite can **shout a message**, and every sprite listening for that message reacts.

Wand



Frog



Tree



The wand has no idea the frog and the tree exist. It just shouts. That's the clever part.

Words to know

Word	What it means
broadcast	Shout a message to every sprite
when I receive	Listen for a message and react to it
state	What "mode" the game is in — title screen, playing, game over
stop all	Stop absolutely everything

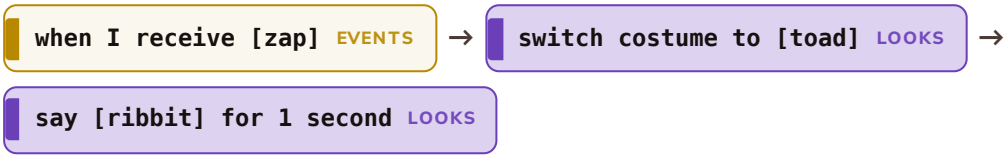
> Message names are typed by hand, so **typos are today's biggest bug**. Always pick the message from the dropdown rather than making a new one by accident.

Class activity — Magic Wand

Wand



Frog



Tree



Now add a second message, `undo`, on a key press, that puts everything back.

Your project — lives, game over, and a start screen

Step 1 — Lives - [ ] Variable `lives`, `set lives to 3` at the start - [ ] Bad catch → `change lives by -1` - [ ] A script that watches for `lives` running out and broadcasts `gameover`



Step 2 — Game over - [ ] A "Game Over" backdrop - [ ] Stage: `when I receive [gameover]` → `switch backdrop to [Game Over]` - [ ] Falling objects: `when I receive [gameover]` → `hide`

Step 3 — Title screen *(if there's time — otherwise it's homework)* - [ ] A "Title" backdrop with your game's name - [ ] `when [s] key pressed` → `broadcast [startgame]` - [ ] Move your game's start code from `when green flag clicked` to `when I receive [startgame]`

- [ ] **Save it**

Homework

1. A way to win. Pick a target score. Broadcast a `youwin` message when the player gets there, with its own backdrop.

2. Restart. Add a way to play again without clicking the green flag — a key or a clickable button that resets your variables and broadcasts `startgame`.

Bonus challenges

1. A "3 — 2 — 1 — GO!" countdown between the title screen and the game starting.
2. **Two-Scene Story.** Two backdrops, two sprites, one story that runs itself — use broadcasts to move from one scene to the next.
3. A boss object that only appears once your score passes 20.
4. Try `broadcast and wait` instead of `broadcast`. What's different? *(Hint: it doesn't carry on until every listener has finished.)*

If something breaks

- **Broadcast does nothing?** Check the message name in the dropdown. There are probably two similar ones.
- **Sprites still visible after game over?** `stop all` ran before they could hide. Hide first, or add a short `wait` before stopping.
- **Sprites invisible when you restart?** They were hidden and never `show`n again.
- **Lives goes to -1, -2, -3?** Use `< 1` instead of `= 0`. Exact checks miss the moment.

Stuck? Put your hand up — that is what we are here for. And always save before you close the lid.